

AI YOGI PROJECT

*Presented by:
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PROBLEMS

01

THERMAL IMAGE LANDMARK AND AREA
PREDICTION

02

EYE MOVEMENT TRACKING

03

YOGA POSE DETECTION





SOLUTION PROCESS

01

DEVELOPED A MODEL USING PYTHON TO FIRST DETECT THE THERMAL IMAGE AND THEN ANNOTATE IT BY PUTTING LANDMARKS AND ALSO PREDICT AREAS WHEREVER REQUIRED

02

DEVELOPED A MODEL USING PYTHON AND OPENCV TO FIRST DETECT THE SPECIFIC FRAME SIZE IN THE INPUT AND MARKS THE IRIS, AND THEN DETECT AND REPRESENT THE MOTION USING A GRAY-SCALE IMAGE

03

DEVELOPED A MODEL USING PYTHON AND OPENCV TO DETECT THE POSTURE OF THE BODY AS INPUT AND MARK BODY PARTS AS POINTS AND DETECT THE ANGLES AND POSITION OF BODY PARTS



FRAMEWORKS AND LIBRARIES

Front End

- Dash, Tailwind CSS

Back End

- Flask, Next.js, Express.js

TO DEVELOP THE
SOFTWARE

OpenCV

Python

Dlib's face detection
Models

TO BUILD THE ML
MODELS



CONTRIBUTION OF THE TEAM

01 ARITRA RAY

- Contributed on the FRONT END of the Relaxation system
- Successfully integrated all the ML models (Pose detection, Eye motion tracking, Thermal image)

02 APARNA PANDEY

- Worked on the THERMAL IMAGING MODEL for thermography and to detect areas temperature like area above the eyes, below lips, on the cheeks etc.

03 BHANDARI SHIKHARA

- Worked on the posture detection and recognition of the input data using OpenCV to detect any bodily motion in the input data.

04 BHAVYA CHARITHA

- Worked on eye motion tracking to detect any changes in the position of iris in OpenCV
- The model is integrated with the software to show the grayscale output for a given input for a particular frame size

05 SIDDHARTH S

- Worked on the backend of the relaxation system using MySQL – database creation, connection, etc.



IMPROVEMENT IN OUR SKILLS THROUGH THE PROJECT



Learned Cutting Edge Technologies

We learned and implemented programming languages, frameworks and libraries which are industry standard. It will help us to build other real time projects in future.



Continuous Improvement and Error Handling

Learning from mistakes is one of the important things we learned during our this project. Initially we tried different types modules and frameworks. But after facing errors in integration we decided to use other frameworks which helped us to handle those errors.

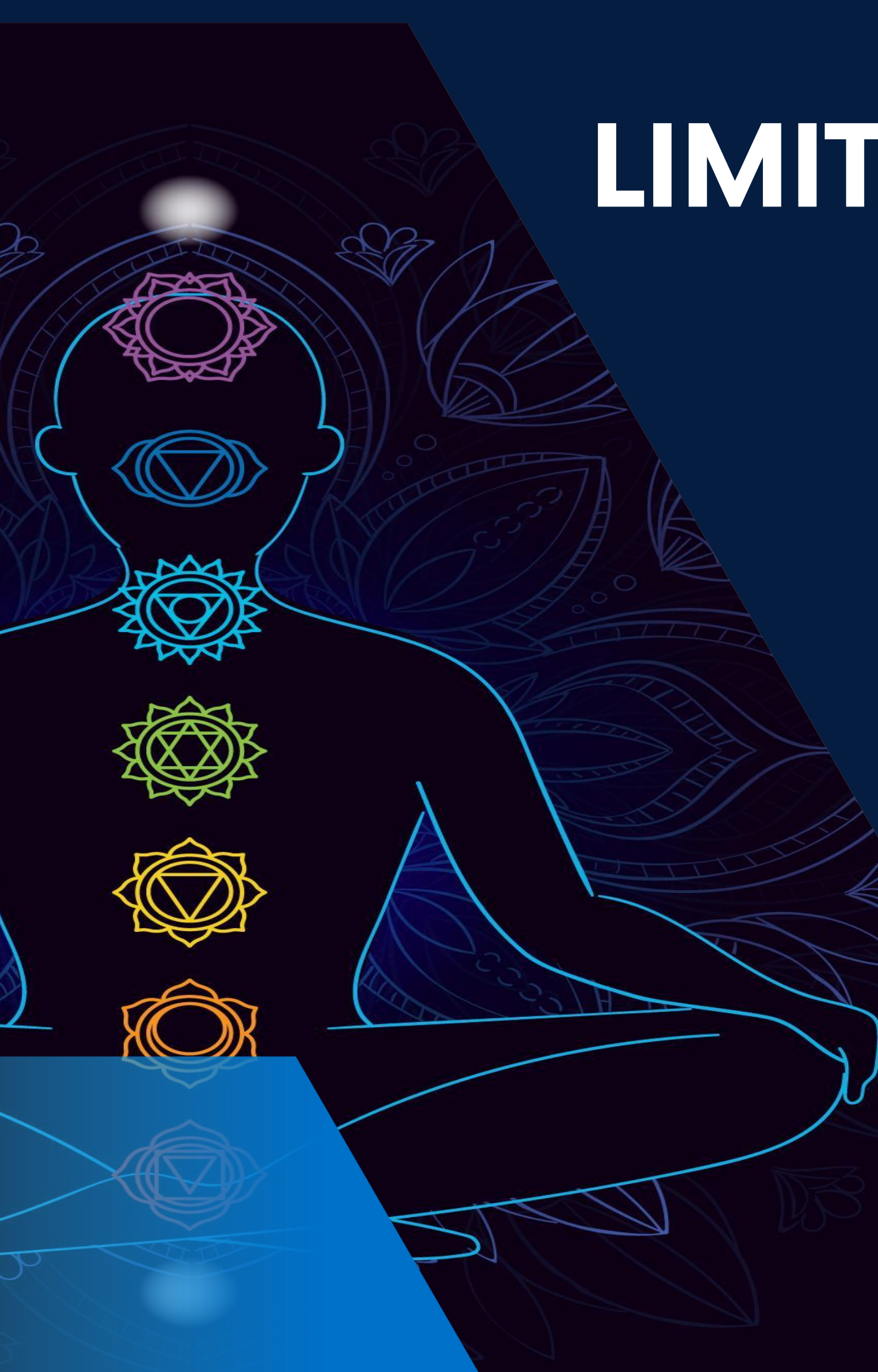


Group work

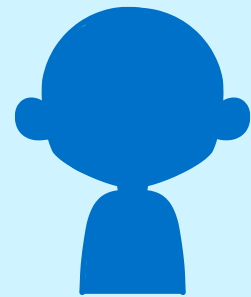
One of the key factor to succeed in a project is to work as a team. We learned this unmentioned skill to excel

LIMITATIONS OF OUR MODEL

- Real-time implementation of complex features like thermal imaging and eye detection is a challenge for our model, making offline analysis using videos from local devices or backend dataframes more practical.
- Images must be resized to the dimensions the model was trained on, such as 464x348, for compatibility with the thermal imaging model.
- Input must allow a specific dimension 1280x720 in pixel for eye motion tracking and grayscale display of the iris motion.
- Use of MySQL for video stream storage is not practical due to the storage as large binary object files, which makes retrieval and storage inefficient



FUTURE SCOPE



THERMAL IMAGE

To make the model flexible such that it can also predict for the videos recorded using thermal imaging cameras.



EYE MOTION TRACKING

The model can be extended to record the count of number of times movement is detected and identify coordinate changes



POSTURE DETECTION

Detection of angles at joints in the posture can be worked upon in addition to the already present other angles

TIMELIN

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January

In this month we familiarized ourselves with the frameworks and libraries that were need and learned from various courses online.

We did the debugging of the existing models and worked on improvement in the model according to the need of this project.

February

March

We worked on the integration of the model in the software along with the all the final touch that were needed to run the model smoothly.

We worked on the presentation part through slides and video to represent our model.

April

VIDEO LINK

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**Thanks For Your
Valuable Time !!**



RESOURCE PAGE

